

Question			Marking details		Marks available				
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Valid method for obtaining gradient (implied if correct) (1) Max = ~0.021 or min = ~0.017 (1) [OK to give 21.0 & 17.5 mT A ⁻¹] Mean correct (~0.019) (1) Uncertainty = 8.5 % – 11% (1) [no s.f. penalty]		4		4	4	4
	(b)		Gradient = $\mu_0 n$ used or implied (n =15 200) (1) N = 3 780 (1) N = 3 800 ± 400 (1) [allow 1 or 2 s.f. in uncertainty 1 more s.f. in N]			3	3	3	3
	(c)	(i)	5 000 is outside uncertainty range [or equiv.] (1) Straight line (1) Through all error bars (1) Lines straddle the origin (1) [allow ...through origin]			4	4		4
		(ii)	Flux density least accurate/probe not in centre [end effect] /not aligned [90° to axis](1) Get better Hall probe with higher [smaller / better] resolution / put in centre/align properly (1) [not: repeat readings / better apparatus]			2	2		2
			Question 4 total	0	4	9	13	7	13